

Ali Imanifard

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Summary

Ranked **1st** out of **100+** students in my M.Sc. cohort for two consecutive years and in my final 18 months of B.Sc. study, achieving a **GPA of 20.00/20.00**, with top performance throughout. Authored 16+ publications focusing on **Human-Centered AI** and **Educational Technology**, including an **IEEE Best Paper Award** and an **IEEE Selected Paper Award**; placed 1st out of 150 submissions at IEEE SGC 2024. Served three years as Research and Teaching Assistant dedicated to developing **pedagogical tools** and adaptive learning frameworks. Committed to advancing scholarly research in **AI-driven education** and **student well-being** at the Ph.D. level.

Education

- M.Sc.** Computer Engineering, Software, Khatam University, Tehran, Iran 2023 – 2025
 University ranked 6th among Iran's 269 non-governmental universities by the Islamic World Science Citation Center (ISC) in 2025. Grad. date: Sept. 2025
- Thesis: *Human-Centered Artificial Intelligence for Enhancing Educational Communication and Student Well-being*, Grade: Excellent 20.00/20.00
 - GPA: **20.00/20.00**, 4.0 (out of 4.0) — *full marks in every course*
- B.Sc.** Computer Engineering, University of Eyvanekey, Semnan, Iran 2019 – 2023
 University ranked 12th among Iran's 269 non-governmental universities by the ISC in 2025 and 1st by the Ministry of Science, Research and Technology in 2014.
- GPA: 18.25/20.00, 3.78 (out of 4.0)
 - Final 18 months GPA: 19.34/20.00, 3.91 (out of 4.0)

Publications

Journal Articles

1. **A. Imanifard**, B. Majidi, and A. Shamisa, "SmartGridAgent: An Educational Framework for Reliable Digital Twin-Based Smart Grid Workforce Training with Locally Hosted LLMs," *Smart Grids and Sustainable Energy*, Vol. 10, no. 2, p. 41, May 2025, doi: [10.1007/s40866-025-00274-0](https://doi.org/10.1007/s40866-025-00274-0) [🔗](#).
2. M. Imani, **A. Imanifard**, B. Majidi, and A. Shamisa, "Dashboard-Driven Machine Learning Analytics and Conceptual LLM Simulations for IIoT Education in Smart Steel Manufacturing," *International Journal of Web Research*, Vol. 8, no. 3, p. 15-31, Summer 2025, doi: [10.22133/ijwr.2025.521822.1280](https://doi.org/10.22133/ijwr.2025.521822.1280) [🔗](#).
3. M. Imani, **A. Imanifard**, B. Majidi, and A. Shamisa, "AI-Integrated Vocational Education and Workforce Reskilling for Steel Industry 4.0," *International Journal of Research and Technology in Electrical Industry*, Vol. 4, no. 2, July 2025, doi: [10.48308/ijrtei.2025.239941.1088](https://doi.org/10.48308/ijrtei.2025.239941.1088) [🔗](#).
4. R. Abniki, **A. Imanifard**, and B. Majidi, "Improving inclusiveness of AI tutoring platform's user experience for marginalized students," *In Revision in International Journal of Web Research*.
5. **A. Imanifard**, E. Akhtarkavan, and B. Majidi, "Preserving Intellectual Property in AI-Generated Student Artworks via A₈ Lattice Vector Quantization in Visual Arts Education," *Under review for Multimedia Systems*.
6. **A. Imanifard**, B. Majidi, and M.T. Manzuri, "Modeling Students' Emotionally Motivated Haptic Behaviors to Address Touchscreen Usability in Digital Education," *Under review for Scientia Iranica*.
7. **A. Imanifard**, M. Imani, and B. Majidi, "RecomPath: An LLM-Empowered Multi-Agent Framework for Intelligent Academic Advising," *Under review for Research in Higher Education*.

8. **A. Imanifard** and B. Majidi, "Agentic experiential learning for climate driven disaster management training," *Under review for Education and Information Technologies*.
9. M. Imani, **A. Imanifard**, B. Majidi, and A. Shamisa, "A Predictive Modeling Framework for Resource Efficiency in Industrial Process Education," *Under review for Sustainable Energy, Grids and Networks*.
10. M. Akbari, **A. Imanifard**, A. Shokri, A. Badamchi, R. Badamchi, and B. Majidi, "Immersive Virtual Reality Wildfire Simulation with Digital Twin Technology for Emergency Training," *Under review for Technology, Knowledge and Learning*.
11. **A. Imanifard**, M. Haji Abbasi, and B. Majidi, "LifeSkillTutorAgent: Human-Centered Multi-Agent AI for Learning and Well-Being in Education," (*In Preparation — Analysis and Drafting*).

Conference Papers

1. R. Abniki, **A. Imanifard**, and B. Majidi, "Mitigation of User Experience Bias in LLM-Driven Educational Web Services for Women and Neurodiverse Learners," in *2026 12th International Conference on Web Research (ICWR)*, 2026: *IEEE Xplore*, doi: <https://doi.org/10.1109/ICWR69602.2026.11513353>.
2. R. Abniki, **A. Imanifard**, and B. Majidi, "Fostering Collaborative Engagement Between Underrepresented Learners and AI Educator," in *2026 International Biennial Conference on Advances in Artificial Intelligence and Data Science (IBAAIDS)*, 2026: *IEEE Xplore*, Bushehr, Iran, pp. 1-5, doi: [10.1109/IBAAIDS66771.2026.11567407](https://doi.org/10.1109/IBAAIDS66771.2026.11567407).
3. R. Abniki, **A. Imanifard**, and B. Majidi, "Evaluation of Gender and Neurodiversity Bias in Generative AI-Based Education and Assessment," in *2026 International Biennial Conference on Advances in Artificial Intelligence and Data Science (IBAAIDS)*, 2026: *IEEE Xplore*, Bushehr, Iran, pp. 1-5, doi: [10.1109/IBAAIDS66771.2026.11567412](https://doi.org/10.1109/IBAAIDS66771.2026.11567412).
4. A. Danesh, **A. Imanifard**, B. Majidi, and A. Shamisa, "Collaborative Agentic IIOT Vocational Education using Language Models for Power Quality Management," in *2025 15th International Smart Grids Conference (SGC)*, 2025: *IEEE Xplore*, Shiraz, Iran, pp. 1-7, doi: [10.1109/SGC69320.2025.11372356](https://doi.org/10.1109/SGC69320.2025.11372356).
5. **A. Imanifard**, B. Majidi, and A. Shamisa, "Trustworthy Large Language Model Prompt Engineering for Risk-Free Smart Grid Management Education," in *2024 14th International Smart Grids Conference (SGC)*, 31 Dec. 2024-1 Jan. 2025, 2025: *IEEE Xplore*, pp. 1-6, doi: [10.1109/SGC64640.2024.10983889](https://doi.org/10.1109/SGC64640.2024.10983889). **[Best Paper Award]**
6. M. Imani, **A. Imanifard**, B. Majidi, and A. Shamisa, "Applications of Large Language Models in Industrial Internet of Things Education," in *2025 11th International Conference on Web Research (ICWR)*, 2025: *IEEE Xplore*, doi: [10.1109/ICWR65219.2025.11006194](https://doi.org/10.1109/ICWR65219.2025.11006194). **[Selected Paper Award]**
7. **A. Imanifard** and B. Majidi, "Effective Android Malware Detection using Emotional Reactions to Sudden Events," in *2024 10th International Conference on Web Research (ICWR)*, 2024: *IEEE Xplore*, pp. 390-396, doi: [10.1109/ICWR61162.2024.10533349](https://doi.org/10.1109/ICWR61162.2024.10533349).

Research Interests

- **Human-Computer Interaction (HCI):** Focus on interpretability and user-adaptive interfaces for digital health and educational wellness.
- **Educational AI Agents:** Investigating multi-agent systems and LLM-based assistants to support personalized learning and student success.
- **Natural Language Processing (NLP):** Studying persuasive computing and dynamic personalized learning profiles to encourage positive behavioral changes in learners.
- **Responsible AI in Pedagogy:** Developing frameworks for the ethical integration of LLMs within educational technology and HCI for development.

Honors & Awards

- **Canada Impact+ Research Training Award (Doctoral)**, Government of Canada, 2026.
 - Prestigious federal Tri-Agency initiative jointly funded by:
 - the Natural Sciences and Engineering Research Council (NSERC)
 - the Canadian Institutes of Health Research (CIHR)
 - the Social Sciences and Humanities Research Council (SSHRC)
 - Administered directly by NSERC to support elite doctoral researchers driving high-impact scientific innovation.
 - Valued at a total of **CAD \$120,000**.
- **Best Paper Award**
 - The 14th *International Smart Grids Conference, IEEE*, 2025.
 - Ranked 1st among 150 papers
 - Received perfect score (100/100) from all reviewers
- **Selected Paper Award**
 - The 11th *International Conference on Web Research (ICWR), IEEE*, 2025.
 - Awarded for its outstanding contribution to the field of Web Research
- **Outstanding Graduate Student Award**, Khatam University, 2025.
- **Outstanding Graduate Researcher Award**, Khatam University, 2025.
- **Ranked 1st in Cumulative GPA & Academic Performance** among all M.Sc. graduates, Khatam University, 2025.
- **Merit-based Scholarship for Outstanding Academic Achievement**, Master's Degree, Khatam University, 2023–2025.
 - Awarded a partial tuition exemption for outstanding academic achievement.
- **Ranked 2nd in Cumulative GPA & Academic Performance** among all B.Sc. graduates, University of Eyvanekey, 2023.
 - Ranked 1st during the final 18 months of the program
- **Outstanding Undergraduate Student**, University of Eyvanekey, 2020–2023.
- **Merit-based Scholarship for Outstanding Academic Achievement**, Bachelor's Degree, University of Eyvanekey, 2021–2023.
 - Awarded a partial tuition exemption for outstanding academic achievement.

Teaching & Research Experiences

Graduate Research Assistant

• Smart Digital Reality Lab

- **Director:** Prof. Dr. Babak Majidi
- **Contributed to 16 publications** focusing on applied human-centered AI by conducting comprehensive literature reviews, synthesizing key findings, and authoring manuscript sections.
- **Designed and implemented data-collection protocols**, including surveys, questionnaires, and experimental setups, gathering qualitative and quantitative data across laboratory studies.
- **Developed research methodologies and hypotheses**, co-designing over a dozen experiments and surveys to rigorously test research questions.
- **Analyzed complex datasets** using statistical and computational techniques, presenting actionable insights that informed iterative study designs.
- **Maintained laboratory and field operations**, ensuring meticulous documentation of procedures, results, and compliance with research standards.
- **Prepared manuscripts, presentations, and posters**, proofreading and editing for clarity, cohesion, and adherence to journal and funding guidelines.

Khatam University
2023 – present

Graduate Teaching Assistant

• Software Testing

- Prof. Dr. Seyed-Hasan Mirian-Hosseiniabadi
- **Mentored student groups** in designing and executing academic projects, providing technical guidance and feedback throughout the course.
- **Delivered a video lecture on JUnit**, hosted on the university's Moodle LMS, as a reusable learning resource for students.
- Provided **one-on-one tutoring** and academic guidance to **30 students** by developing, administering, and grading assignments, quizzes, and exams while offering constructive feedback to enhance their understanding and performance.
- Fostered a positive and engaging learning environment by holding regular office hours, effectively communicating course updates, and addressing student questions and concerns regarding the curriculum and assignments.

Khatam University
2025 – present

• **Advanced Software Engineering**

- Prof. Dr. Seyed-Hasan Mirian-Hosseiniabadi
- Delivered individualized tutoring and comprehensive academic support to students by managing the **grading process for a class of 30 students**, developing and grading assignments, quizzes, and exams, and providing constructive feedback on their performance.
- Fostered a collaborative and engaging learning environment by mentoring students, addressing their inquiries and concerns regarding course material and assignments, and holding regular office hours for personalized support.
- Effectively communicated course updates and essential information, consistently earning positive student feedback for my contributions as a teaching assistant.

Khatam University
2024 – present

• **Advanced Artificial Intelligence**

- Prof. Dr. Babak Majidi
- Led student mentorship on **LLM Agents** and generative AI workflows, introducing **chain-of-thought prompting** techniques and coaching learners to build custom LLM Agents for educational tutoring and automated study-guide generation.
- Guided **70+ students on transformer-based LLMs** (e.g., GPT, BERT) during lectures and virtual meetings, providing real-time feedback on hands-on prompt-engineering exercises.
- Compiled and communicated cutting-edge LLM research trends, including prompting best practices, retrieval-augmented generation, and LLM Agent integration with external APIs, enabling students to critically appraise and apply recent publications.
- Provided one-on-one tutoring in advanced NLP frameworks, locally hosted LLM integration, and ethical AI considerations, resulting in a measurable increase in student project quality.
- Earned consistently outstanding feedback for clarity and engagement in explaining lessons on machine learning, deep learning, and deploying LLMs in real-world applications.

Khatam University
2024 – 2025

• **Statistical Learning Theory**

- Prof. Dr. Shiva Kamkar
- **Delivered lectures** on “An Introduction to the R Programming Language” and “Data Visualization in R with ggplot2 (Beginning to Advanced),” **led weekly discussion** sections, and conducted **13 recitation sessions**, effectively clarifying complex concepts while guiding students through problem-solving and application of statistical learning theory.
- Developed, administered, and graded assignments, quizzes, and exams, and provided constructive feedback to improve student understanding.
- Provided comprehensive academic support by addressing student questions and concerns regarding course material and assignments, thereby contributing to a positive and engaging learning environment.

Khatam University
2024 – 2024

Skills

- **Programming Languages**

- Python, R, Java, C, C++, SQL, Julia, MATLAB

- **LLM & AI Development Environments**

- LLM Agent Frameworks
 - CrewAI, LangChain, Llama Index, OpenAI Agents SDK
- Embeddable Local AI Engines
 - llama.cpp, Ollama, PrivateGPT, GPT4All, Msty, LM Studio, Jan
- LLM-Specific Tools and Techniques
 - Transformers (Hugging Face)
 - Prompt Engineering
 - Few-Shot Prompting, Chain-of-Thought Prompting, Retrieval Augmented Generation (RAG), Generate Knowledge Prompting, Program-Aided Language Models (PALs), Tree of Thoughts (ToT), ReAct, Reflexion, etc.
- NLP Libraries
 - spaCy, NLTK
- General AI Development Frameworks
 - TensorFlow, Keras, PyTorch, scikit-learn
- Visualization Tools
 - Matplotlib, Seaborn, Plotly, ggplot2
- Data Processing Tools
 - Pandas, NumPy, dplyr

- **Engineering Software Skills**

- IDEs: Anaconda, RStudio, PyCharm, Visual Studio Code, IntelliJ IDEA
- Interactive Application Development Tools: Gradio, Streamlit
- Data Science and Analytics Platforms: RapidMiner

- **Academic Writing Tools**

- LaTeX, Overleaf, EndNote, Mendeley, Zotero

- **Core Competencies**

- Git, GitHub
- Jupyter Notebook, Google Colab, Kaggle Notebooks
- MySQL, Microsoft SQL Server, PostgreSQL, MongoDB, SQLite
- Apache Spark, Hadoop
- Microsoft Office, Google Workspace (formerly G suite)
- Canva, Figma

- **English Test Scores**

- TOEFL (iBT): **5.0/6.0 (=100/120)** (overall score)
 - Reading: 5.5 / 6.0
 - Listening: 5.0 / 6.0
 - Writing: 5.0 / 6.0
 - Speaking: 4.0 / 6.0
 - Test Date: July 1st 2026

Voluntary Services

Master's Thesis Guidance and Feedback Facilitator

- Provided comprehensive feedback and full reviews for master's students before their thesis defense, assisting with structure, content, and defense preparation.

Khatam University
2024 – present

Peer Support in Academic Writing and Research

- Provided academic support to university students by reviewing their theses and articles. Assisted with research design, academic writing, and presentation prior to faculty review.

Khatam University
2024 – present

Organizer of Free Research Training Sessions

- Organized and led online sessions on research methodologies, proposal writing, and scientific paper drafting, providing valuable resources and training to participants at no cost.

Khatam University
2024 – present

Educational Case Advisor

- Assisted master's students in selecting thesis topics, advising on feasibility and research potential to align with academic and career aspirations.

Khatam University
2024 – present

Academic Mentor and Study Group Leader

- Led undergraduate study groups, managing project timelines, task delegation, and progress reporting. Provided leadership, project management, and mentorship to facilitate successful project completion.

Khatam University
2024 – present

Selected Academic Projects

LLM-Empowered Multi-Agent Framework for Academic and Career Path Optimization Using Newly Collected GradCafé Dataset, 2025

- Developed a multi-agent system utilizing a newly scraped, comprehensive GradCafé dataset to predict university admission chances, recommend resume improvements, and provide personalized academic and career guidance powered by LLM.

Repository available upon
publication acceptance

Emotion-Aware Modeling of Student Experience in E-Learning via Touchscreen Response Analysis, 2025

- Developed an emotion-aware framework to analyze touchscreen interactions to improve user experience (UX) and student well-being in mobile e-learning environments.

Repository available upon
publication acceptance

LLM-Based Disaster Management Multi-Agent System, 2025

- Developed a pedagogical simulation using LLMs to help students practice crisis management theory, focusing on the educational application of agentic workflows.

[github.com/Alilmanifard/
LLM-Disaster-Management-
Agents](https://github.com/Alilmanifard/LLM-Disaster-Management-Agents) 

SmartGridAgent Educational Simulation for Power Systems Concepts Using Locally Hosted LLMs, 2025

- Designed a digital twin-based training environment intended for workforce education, allowing students to learn the principles of complex system concepts in a simulated academic setting.

[github.com/Alilmanifard/
SmartGridAgent-LocalLLM](https://github.com/Alilmanifard/SmartGridAgent-LocalLLM) 

Vision Transformer (ViT)-based Evaluation and Stable Diffusion Metadata Generator for Academic Image Datasets, 2025

- Developed a metadata generation pipeline for Stable Diffusion datasets and designed an evaluation framework using ViT and SSIM to assess image quality and metadata retrieval performance by comparing original and reconstructed images.

[github.com/Alilmanifard/
Stable-Diffusion-metadata-
generator](https://github.com/Alilmanifard/Stable-Diffusion-metadata-generator) 

Analysis of Socioeconomic Factors on Student Academic Performance and Outcomes: A Statistical Learning Theory Approach, 2024

- Investigated the impact of socioeconomic factors and student academic choices on educational attainment using advanced Statistical Learning Theory methods. Performed data preprocessing, feature engineering, outlier detection, and statistical modeling to analyze student performance and graduation likelihood.

github.com/Alilmanifard/Student-Performance-Analysis-SLT 

Enhancing Student Engagement in Software Testing Education Using JUnit, 2024

- Designed and implemented superhero-themed JUnit lecture and interactive materials to improve student understanding of software testing concepts. Implemented in Java using NetBeans IDE with an emphasis on practical unit testing skills.

github.com/Alilmanifard/junit-engagement-software-testing 

Distributed Natural Join Implementation Using PySpark and MapReduce, 2023

- Developed a PySpark-based system to execute a natural join on distributed CSV datasets using RDD transformations, simulating MapReduce logic in a Hadoop environment.

github.com/Alilmanifard/spark-natural-join-mapreduce 

Work Experience

AI Research Engineer

- Commercial Fabric & Consumer Textile Manufacturing
- Develop and implement machine learning models and automated software tools to support internal analytics and business workflow optimization.
- Research and design multi-agent LLM frameworks for academic knowledge synthesis, focusing on generating structured educational content from public scholarly literature.
- Apply statistical and machine learning methods to analyze market data and support internal data-driven decision tools.
- Build recommendation systems for consumer products, applying machine learning techniques to improve personalization and user experience.

Negin Rose Sepahan
Textile Company
2025 – *present*

References

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